

Bowen Li

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EDUCATION

University of Southern California <i>M.S. Industrial and Systems Engineering – Analytics ; GPA:3.61/4.00</i>	Los Angeles, CA Aug 2022 – May 2024
Dalian Polytechnic University <i>B.S. Computer Science and Technology; GPA:3.40/4.00</i>	Dalian, CHN Aug 2018 – May 2022

SKILLS

Programming: Python(Pandas, NumPy, Scikit-learn, LightGBM, Pytorch, OpenCV), R(GLMM, GAM), SQL, HTML/CSS, Shell
ML & Sports Analytics: Bayesian modelling(PyMC), Computer Vision(YOLOX, MixSort, CLIP ReID, RTMPose, YOLOV11)
Data Engineering & Tools: MySQL, MongoDB, GCP, AWS, Linux/UNIX, Git, Postman, Docker, Kubernetes, Streamlit, R Shiny

WORKING EXPERIENCES

Athleo.ai <i>Developer of Basketball Video Auto-Clipping System</i>	Remote Jan 2026 –Present
Pipeline Architecture: Designed an end-to-end auto-clipping workflow for basketball footage using Python, PyTorch, and OpenCV, enabling user-specific clip generation based on selected players in under 30 seconds per game segment, reducing manual editing time by 80%. Detection & Tracking: Applied YOLOX with MixSort to process raw game footage, producing per-frame player IDs and bounding boxes with >85% tracking accuracy, ensuring reliable downstream analysis Player Re-Identification: Implemented CLIP-based embedding comparison with cosine similarity to link player appearances into tracklets, improving clip segmentation accuracy. Modelling & Visualization: Trained Mask R-CNN models for basket, court line, and ball segmentation to deliver precise visual overlays with b-box/segmentation AP >90%, enhancing user viewing experience.	
Cognizant – GOOGLE XR ML Data Collection & Annotation <i>Data Analyst</i>	Los Angeles, CA Jan 2025 – Present
Data Visualization: Designed a Tableau dashboard analyzing 10,000+ survey responses across 20 projects, providing insights on user satisfaction, facilitator performance, and equipment usability. Utilized n-gram word clouds, drill-down analysis, and network diagrams to uncover recurring device issues, leading to hardware improvements and streamlined data collection processes. Automation & Scripting: Automated CSAT survey proctor name updates using Google Sheets, App Script, and SQL, ensuring real-time synchronization and reducing manual workload. Eliminated 5+ hours of manual interventions per week, enhancing data accuracy and operational efficiency across multiple projects. Cross-functional Collaboration: Partnered with hardware engineers, project leads, and research participants to execute high-volume VR headset data-collection sessions for face-tracking research, ensuring >98% session data integrity.	

SPORTS ANALYTICS EXPERIENCE

<u>NBA Draft 2026 – Shot Quality Scouting Portal</u>	Apr 2026 – May 2026
Data Pipeline: Built an end-to-end NCAA shot data pipeline via CBBD API, ingesting shot-by-shot records for 65 draft prospects into SQLite with automated box score, bio, and international stat collection — covering the full data foundation for prospect evaluation. Modeling & CV Pipeline: Trained a LightGBM xPTS model with Empirical Bayes shrinkage to compute PAE/100 per prospect; built a YOLOv11n jump-shot classifier (mAP50 = 0.828) and integrated RTMPose for frame-level joint angle extraction across 20+ clips per player. Scouting Dashboard: Delivered a 4-page Streamlit portal featuring a sortable draft board, per-player dossiers with Plotly radar charts and half-court zone diagrams, head-to-head comparison, and film-based mechanics analysis — enabling end-to-end prospect evaluation from raw shot data to video mechanics.	
<u>Player Development Pitching Dashboard (R Shiny)</u>	Jan 2026 – Feb 2026
Metric & Logic Layer: Engineered four core PD metrics (BB%, K%, SLG%, FPS%) with rolling trend computation and goal evaluation logic, supporting season-baseline vs. configurable recent-window comparisons per pitcher. Coach-Facing Product: Delivered a 4-tab R Shiny portal (Player Overview, Trends, Goals, Report) with metric cards, rolling trend charts, goal progress indicators, and auto-generated 1-page PDF reports — designed to surface actionable pitcher insights in under 30 seconds.	
<u>Catcher Framing Modelling Project - Arizona Diamondbacks (Interview Take-Home)</u>	Nov 2025 – Dec 2025
Modeling: Isolated catcher contribution to ball/strike calls using a hierarchical mixed-effects model with a LightGBM probability baseline (5-fold CV, AUC = 0.98), engineered from spatial and count-context pitch-tracking features. Pipeline: Built a production scoring pipeline that loads the trained model, scores incoming pitch-level data, and auto-generates year-by-catcher framing summaries.	